Examiner reports

Q1.

Here it was insufficient to state the midpoint or the gradient of the perpendicular bisector. These facts needed to be applied in a method leading towards finding the values of c and d.

Many correctly obtained the required gradient equation. Fewer substituted the coordinates of the midpoint into the equation of the bisector. Many instead stated that *C* and *D* lay on the bisector leading to wrong equations. Even where the two correct equations were obtained, poor algebra often led to wrong answers. A common error was to say that since $\frac{d-2}{6-c} = \frac{1}{4}$ then d = 3 and c = 2.

Q7.

(a) (i) Those who began by substituting x = 7 into the equation of AB were usually successful. Another approach was to rearrange the given equation in order to find the gradient and then equate this value to the gradient found using the points (–3, 2) and (7, k). Quite a few attempted to verify the result by more complicated methods, but such attempts rarely had an appropriate conclusion and did not always convince the examiner.

(ii) Apart from the occasional arithmetic error, this was answered well. A few weaker students used an incorrect midpoint formula, usually having minus signs instead of plus signs.

(b) Once more this was answered well. The most successful students rearranged the given equation to make y the subject, but others used the two pairs of coordinates. A few students made sign errors due to the negative coordinates but most students found the correct value of the gradient.

(c) Most students realised the need to find the negative reciprocal of the gradient of AB in order to find the gradient of the perpendicular line. Very few failed to do so. However, not all heeded the question where a specific form of answer was requested. A common incorrect answer was $5x - 3y + 7 = 0$ where students, having obtained the equation $y = \frac{5}{3}x + 7$, neglected to multiply the constant term by 3. Those who used $y = mx + c$ for the equation of the straight line usually made more mistakes than those using the form $y - y_1 = m(x - x_1)$.

(d) There were several completely correct solutions here. The question was easier than on previous papers because students did not have to decide which equations to use to find the coordinates of C. However, amongst the weaker students, sign and arithmetic errors abounded.

Q8.

(a) Students must be aware that, when they are asked to complete a proof, each stage must be clearly and accurately shown. This was not always the case, and careless sign errors and the omission of "= 0" were penalised.

(b) (i) A very common error here, in finding the discriminant, $b^2 - 4ac$, was where students explicitly wrote " $b = 2x + 1$ " instead of "$b = -(2x + 1)$" . Those who did have the correct value of b sometimes wrote b^2 as $-(2x + 1)^2$. Unfortunately, many students did not write the condition for distinct real roots until the very last line which was insufficient, since this was the given answer.

The better solutions began by quoting the condition that two distinct real roots occur when $b^2 - 4ac > 0$ or the appropriate inequality with the discriminant expressed in terms of k .

(ii) It was pleasing to see some completely correct solutions here. Those who chose to factorise the equation, in order to find the critical values, were usually more successful than those using the quadratic equation formula, as $\sqrt{20^2 - 16 \times 9}$ defeated many without the use of a calculator. Students are advised to draw a sketch or sign diagram in order to solve a quadratic inequality. A few thought that the two inequalities, $k < \frac{1}{2}$, $k > \frac{9}{2}$ should be combined and so spoiled their correct solution by writing something like $\frac{1}{2} > k > \frac{9}{2}$ as their final answer, and this was penalised.

Q9.

(a) (i) Almost all candidates began by substituting $x = p$ and $y = p + 2$ into the given equation. However, many made a sign error when multiplying out $-4(p + 2)$ and consequently $p = 13$ was a very common incorrect answer.

(b) Most candidates rearranged the equation to the form $y = mx + c$ and gave the correct gradient. Some used the coordinates of two points on AB, such as (1, 2) and (–3, –1). Some of the weaker candidates chose two values for p, found 2 pairs of incorrect coordinates and then attempted to find the gradient of the line joining these points.

(c) There were two equally common methods used here. Both required the use of the negative reciprocal of the answer to part (b) to provide the relevant gradient and it was pleasing to note that most candidates realised this. One approach was to find the gradient in terms of k and equate it to the relevant gradient and solve; the alternative was to set up an equation using the coordinates of A (or C) and then to substitute the coordinates of the other point to obtain the value of k.

(d) Although most candidates realised the necessity to set up 2 equations in order to eliminate x or y, arithmetic and sign errors abounded and there were many incorrect solutions. Even those who correctly found $7y = -28$, say, often wrote down an incorrect value of y such as $y = -3$ or $y = 4$. A few candidates misread the question and used an incorrect equation for one of the lines.

Q10.

(a) Those candidates who began by finding $\frac{dy}{dx}$ usually found the correct gradient and most gave the equation of the tangent in the required form. A few made arithmetic errors such as $y - 6 = 9x + 9$ leading to $y = 9x + 3$. A few, having found the gradient to be 9, then incorrectly used the negative reciprocal , i.e. $-\frac{1}{9}$. Some candidates did not realise that differentiation was required and attempted to find the gradient using the coordinates of the two points (–1, 6) and (2, 33).

(b) (i) Provided no arithmetic errors were made, particularly evaluating $2^5 = 64$, then the correct value of k was found from the equation of the curve. A few candidates used the tangent equation and although they found that $k = 33$, no credit was given.

(ii) Those who used the equation of the line did not always earn the mark as a concluding statement such as "therefore lies on the tangent" was not made. Some worked backwards and found the gradient using (2, 33). A few candidates confused the requests in parts (i) and (ii) and consequently earned no marks.

(c) (i) The majority of candidates integrated correctly. Most substituted the limits in the correct order but many sign and arithmetic errors were made. A common error was evaluating $\frac{5}{6} - \frac{54}{6}$ as $\frac{49}{6}$ instead of $-\frac{49}{6}$ and the correct value of the integral was rarely seen.

(ii) Very few candidates found the area of a trapezium. Most found the area of a triangle instead. Those who used integration in order to find the area under the straight line were far more successful than those who tried to remember the basic formula from GCSE.

Q11.

In part (a), most candidates were able to complete the square for at least one of the terms. A very common incorrect answer involved $(x - 7)^2$ instead of $(x + 7)^2$. Occasionally a negative answer was given for the square of the radius. Also, $\sqrt{5}$, $\sqrt{25}$ and ± 49 were often seen on the right-hand side of the equation.

In part (b), the coordinates of the centre usually followed through correctly from part (a). Often the correct radius was stated, even though part (a) was incorrect.

In part (c), the centre of the circle usually appeared in the correct quadrant for the candidate's coordinates. However, those who had the correct answers in part (b) did not always draw the correct circle touching the x -axis at –7 and not touching the y -axis.

In part (d)(i), several candidates earned the method mark but the most common error lay in the failure to square (kx) as k^2x^2; most wrote kx^2 and immediately lost the accuracy mark.

In part (d)(ii), candidates seemed to think it necessary to write "= 0" only on the very last line. This does not fulfil the requirements of a proof. The first line should be that equal roots occur when $b^2 - 4ac = 0$ and the "= 0" should continue throughout.

Other common errors occurred in the squaring of $2(k + 7)$, which often became $2(k + 7)^2$. Candidates should check each line of their proof instead of fudging one or more lines of working, as examiners are unlikely to be deceived by the miraculous appearance of the printed answer following incorrect working.

Many did not attempt part (d)(iii). The factorisation of the quadratic expression defeated most candidates; too many are reliant on the quadratic equation formula which often produces large numbers that are difficult to handle without a calculator. The incorrect factors $(4k - 3)(3k + 4)$ were seen as often as the correct ones. Candidates found this factorisation extremely hard and very few picked up these two final marks, but this is a topic that should not be beyond students at this level.

Q12.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the gradient m to be –8. With the answer given, it was not acceptable to just state that the gradient is –8 without some clear indication that this value came from the value of $\frac{dy}{dx}$ when $x = 8$. In part (c), the most common loss of a mark was through failing to simplify $\frac{3}{\left(\frac{8}{3}\right)}$ correctly. The error of integrating $x^{\frac{5}{3}}$ to get a term in the form $kx^{\frac{7}{3}}$ was penalised more heavily.

In part (d), those who found the area of the triangle by solving the equations of OP and AP simultaneously to find the coordinates of P and then using $A = \frac{1}{2}bh$ were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for OP and then 3.2 to 8 for AP. It was also surprising to see a significant minority of candidates finding the lengths of OP and AP and the angle OPA and then using $\frac{1}{2}ab\sin C$ to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

Q13.

In part (a)(i), most students were able to find the correct gradient, although some insisted on writing the gradient as $\frac{4}{3}x$. Unfortunately, quite a few were unable to make y the subject of the equation $4x - 3y = 7$ and, consequently, the incorrect value $-\frac{4}{3}$ was offered as the gradient by a significant number of students; a minority confused the gradient with the y-intercept and gave answers of $\pm\frac{7}{3}$.

In part (a)(ii), the most successful attempts at the equation of the line used an equation of the form $y - y_1 = m(x - x_1)$, as flagged above. Many who tried to use $y = mx + c$ lost marks because they made arithmetic errors when trying to find the value of c. Quite a large number did not appreciate that parallel lines have the same gradient and these usually found the negative reciprocal of their gradient from part (a)(i) to use for the gradient of the parallel line. Although most students found a correct equation, many mistakes were seen when trying to rearrange their equation into the required form with integer coefficients.

In part (b), most students made an attempt at the simultaneous equations, but it was alarming to see how few obtained the correct solution. Poor algebraic skills and carelessness were often evident here; elimination of one of the variables proved to be the most successful approach; those using substitution were usually unable to cope with the fractions and negative signs. Some students did not read the question carefully, and no credit was given to those who used the wrong pair of equations.

In part (c), once again, poor algebra was the main problem. Many of those students who successfully wrote $4(k - 2) - 3(2k - 3) = 7$ were unable to solve the equation for k correctly.
Others scored no marks for substituting into the wrong equation.

Q14.

In part (a)(i), the coefficient of x, being an odd number, caused problems for some when completing the square; most students could not square 1.5 without a calculator; those who used fractions could not always simplify $5 - \frac{9}{4}$, and for those using decimals $-1.25 + 5 = 3.75$ was commonly seen.

In part (a)(ii), the equation of the line of symmetry was not usually stated correctly, with several writing down the equation of a curve rather than a straight line; the incorrect equation $y = \frac{3}{2}$ was seen far too often, whereas others simply wrote down numbers such as $\frac{3}{2}$ or $\frac{11}{4}$; this topic was not well understood by many students.

In part (b)(i), most attempts earned the method mark for eliminating y and collecting like terms, but, surprisingly, not all were able to solve the resulting quadratic equation. A common incorrect solution to $x^2 - 4x = 0$ was $x = \pm 2$, whereas others obtained $y = x^2 - 4x$ from the two equations and made no progress.

Integration of polynomials is well drilled and part (b)(ii) earned full marks for weak and strong students alike. There were few errors seen but the most common involved the constant term; some kept the term as 5 and others omitted the final term possibly confusing differentiation with integration.

In part (b)(iii), most students realised the need to substitute appropriate limits but, without a calculator, arithmetic errors were abundant; 4^3 was often evaluated as 48 and the fractions and minus sign caused problems for many students who could not simplify $\frac{64}{3}$ –

24 + 20.
Those who were unable to complete part (b)(i) often chose the value 5 as their upper limit, possibly since the coordinates of A were (0, 5), but no credit was given for this. However, those who had the solution x = 2 from their earlier work were given a method mark for using this value as their upper limit. A large number of students presented the value of the definite integral as their final answer to the area of the shaded region instead of considering the difference of the area of the appropriate trapezium and the value of the integral.

Q15.

In part (a)(i), the idea of y being increasing when $\frac{dy}{dx} > 0$ seemed unfamiliar to many students, who seemed content to substitute a few numerical values for x into the expression for $\frac{dy}{dx}$ rather than tackling the three-line proof. Consequently, very few scored full marks on this part.

Many students ignored the inequality signs throughout part (a)(ii). It was pleasing to see many students attempting to factorise the quadratic, often successfully, rather than relying on the formula. Perhaps more students would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. There were many correct solutions, but students must realise that $x < 2$, $x > \frac{4}{3}$ is not the same as $\frac{4}{3} < x < 2$

In part (b)(i), most students attempted to evaluate $\frac{dy}{dx}$ when x = 2. Not all were able to cope with the evaluation of $-6(2)^2$, which they needed to do in order to convince examiners that $\frac{dy}{dx}$ was indeed equal to zero. Once more, some failed to write a concluding statement or referred to a stationary point rather than the fact that the tangent was parallel to the x-axis.

In part (b)(ii), it was pleasing to see some fully correct solutions. Most recognised that the equation of the tangent at P was y = 3, but it was disappointing to see a number of students write $y - 3 = 9(x - 2)$ followed by $y = x + 1$ as their tangent equation. In finding the equation of the normal at Q, some, having correctly found that $\frac{dy}{dx} = 10$ when $x = 3$, used the value 10 instead of $-\frac{1}{10}$ for the gradient of the normal. Arithmetic errors were very common by those who used $y = mx + c$ when finding the value of c; typically $y = \frac{x}{10} + \frac{7}{10}$ was seen instead of $y = \frac{x}{10} - \frac{13}{10}$. Curiously, quite a few students who had previously obtained y = 3 as the equation of the tangent, and consequently the correct y-coordinate of the point R, then substituted the correct value of x into their incorrect equation for the normal and found a different, and now incorrect value for the y-coordinate of R.

Q16.

In part (a)(i) Although the majority of candidates found the correct gradient, some lost the – sign and others wrote their answer as $-\frac{3}{2}x$. Those who used two correct points often made a sign error in simplifying their expression for the gradient.

In part (a)(ii) it was disconcerting to see so many candidates finding the equation of a line perpendicular to AB rather than the equation of a line parallel to AB. Even having the correct line equation did not mean that full marks were earned; quite a few found the value of x when $y = 0$ and others who correctly obtained the equation $3x + 2y + 8 = 0$ quoted the intercept as being at $y = -8$.

Part (b) was generally well done, although eliminating y and getting as far as $5x = -5$ did not always result in the correct solution; final answers of $x = -5$ or $x = 1$ were seen far too often.

In part (c) the relevant distance formula expression often had a sign error in at least one bracket or was equated to 5 instead of 25. However it was good to see a sizeable number of fully correct solutions. Some candidates used a diagram and a vector approach and wrote down the correct values of k.

Q17.

In part (a)(i) the negative coefficient of x^2 confused many candidates. The correct answer was rarely seen. Those who chose to work initially with the negative of the given expression, even successfully finding $(x + 5)^2 - 29$, often wrote their final answer as $29 - (x - 5)^2$. Candidates need to practise completing the square for expressions of this type even though it has not previously been tested very often.

In (a)(ii) too many had no idea how the line of symmetry related to the previous part. Many candidates just wrote down the maximum point or the equation of the parabola.

In part (b)(i) although the majority of candidates realised that they should equate the two equations, numerous algebraic errors appeared before the final line and even this final equation was sometimes incorrect.

In part (b)(ii) even some able students were unable to cope with squaring $2(2k + 5)$ in order to find an expression for the discriminant. Again, poor algebra and fudging abounded in order to obtain the printed inequality. To earn all the marks it was essential to have stated the condition
$b^2 - 4ac > 0$ early in their working.

In (b)(iii) Only correct factors or quadratic formula earned the first mark; candidates at this level should all be capable of this, particularly with 29 being a prime number. Unfortunately, even with correct factors, some lost the minus sign on one of their critical values. Many failed to draw the appropriate sketch or sign diagram for determining the correct solution, and a few lost the final mark by trying to combine the two correct inequalities thus producing an intransitive statement
such as $-\frac{29}{4} > k > -1$.

Q18.

In part (a) most candidates were able to find the correct gradient; however, some were unable to make y the subject of the equation $7x + 3y = 13$.

In (b)(i) the most successful attempts at the equation of the line used an equation of the form
$y - y_1 = m(x - x_1)$ as flagged above. Many who tried to use $y = mx + c$ lost a mark because they made arithmetic errors when trying to find the value of c. Unfortunately, some candidates used the gradient of the line perpendicular to AB.

In part (b)(ii) many candidates used a method based on differences of coordinates or simply wrote down the correct coordinates of A, but others misunderstood the request and actually found the mid-point of A and the given point.

Part (c) Most candidates made an attempt at the simultaneous equations but it was alarming to see how few obtained the correct solutions. Poor algebraic skills and carelessness were often evident here; elimination of one of the variables proved to be the most successful approach; those using substitution were usually unable to cope with the fractions. No credit was given to candidates who used the wrong pair of equations.

Q19.

In part (a) most candidates found the correct values of a and b, but correct values for k were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side.

In (b) some used an approach based on Pythagoras, but the majority of candidates used an algebraic method, substituting $y = 0$ into their circle equation and solving the resulting quadratic equation in x. Although there were a number who made sign errors, many candidates produced the correct values of x. Nevertheless, common mistakes such as putting $x = 0$ or $(y + 8) = 0$, instead of $y = 0$, resulted in no marks being earned.

In part (c) many candidates assumed that the gradient of CA was 2.5 instead of finding the negative reciprocal and many who did have a correct equation were unable to produce an equation with integer coefficients as requested. A sketch might have helped some to have seen the correct configuration and perpendicularity.

In (d)(i) most candidates realised the need to substitute $y = 2x + 1$ into one or other form of their circle equation but algebraic errors abounded as they tried to produce the displayed quadratic equation. Those who expanded the circle equation first were generally more successful; others who did not have 100 on the right hand side of their circle equation failed to recognise this error when they failed to produce the printed answer for the equation.
Candidates should be strongly discouraged from starting each new line with an equals sign when producing an algebraic proof of this type that involves equations.

In (d)(ii) those who completed the square failed to score full marks if they simply wrote $x + 3 = \sqrt{11}$ before concluding that $x = -3 \pm \sqrt{11}$. Those using the quadratic equation formula often struggled to simplify $\frac{-6 \pm \sqrt{44}}{2}$.

Q20.

Many candidates did not have enough room to complete this question in the space available, often because of over–elaborating their answer to part (b)(ii).

Part (a) was answered correctly in the majority of cases.

Part (b)(i) was also done well, with only a small number making slips with signs, or square-rooting when they should have squared.
Part (b)(ii) had only 1 mark allocated and a request to "write down" the required equation. Although approximately 25% of the candidates did write down the correct equation, $x = 4$, of the normal at the maximum point, a far more popular but wrong answer was $y = 8$. Many other candidates produced half a page of working to reach an equation of a line which was neither vertical nor horizontal.

It was pleasing to see a much better response in part (c)(i) with many more finding the correct equation of the normal at P, although many failed to reduce it to a form with positive integers a, b

and, in particular, c. It was not uncommon to see the final answer left as '$2x + 3y = \frac{99}{4}$', just one
step away from the correct answer '$8x + 12y = 99$'. In general, only the more able candidates, who made good use of the given diagram to show that the normal at M was vertical, gained credit for their answers to part (c)(ii) of the question.

Q21.

Part (a) Many candidates were unable to make y the subject of the equation $2x + 3y = 14$ and, as a result, many incorrect answers for the gradient were seen. Those who tried to use two points on the line to find the gradient were rarely successful.

Part (b)(i) Those candidates who obtained a value for the gradient in part (a) were usually aware that the line DC had the same gradient. Those using $y = mx + c$ often made errors when finding the value of c, whereas those writing down an equation of the form $y - y_1 = m(x - x_1)$ usually scored full marks.

Part (b)(ii) Most candidates realised that the product of the gradients of perpendicular lines should be –1 and credit was given for using this result together with their answer from part (a). Careless arithmetic prevented many from obtaining the final equation in the given form with integer coefficients.

Part (c) Although many correct answers for the coordinates of B were seen, the simultaneous equations defeated a large number of candidates. No credit was given for mistakenly using their equation from part (b)(i) or part (b)(ii) instead of the correct equation for AB, clearly printed below the diagram. Many did not recognize the need to use the equation of AB at all. It was common to see $x = 0$ or $y = 0$ substituted into the equation for BC and then solved to obtain the other coordinate.

Q22.

In part (a), many candidates applied at least one correct formula, but it was not uncommon to see $S_n = \frac{n}{2}(a + l)$ quoted and used as though l, the last term, was 1, or l not replaced by $a + (n - 1)d$. A significant number of average candidates attempted a verification or just used '38 = a + 24 ×1.5' to get 'a = 2' and then applied the same general equation to get the circular argument '38 = 2 + 24d so $d = \frac{38 - 2}{24} = 1.5$'.

Some better candidates correctly obtained the two equations '38 = a + 24d' and '1250 = 20(2a + 39d)' but then just wrote down 'd =1.5, a = 2'. As 'd = 1.5' is printed in the question, candidates were required to show sufficient working for the solution of these simultaneous equations before the final two marks could be awarded. The most common error in part (b) was candidates linking S_n to 100 rather than using $u_n < 100$. However, it was pleasing to see many of those candidates who could not answer part (a) using the printed value for d to answer part (b).

Q23.

In part (a)(i) many candidates were unable to make y the subject of the equation $3x + 5y = 11$ and, as a result, many incorrect answers for the gradient were seen. Those who tried to use two points on the line to find the gradient were rarely successful.

In part (a)(ii) most candidates realised that the product of the gradients of perpendicular lines should be –1 and credit was given for using this result together with their answer from part(a)(i).

Although many correct answers for the coordinates of C were seen in part (b)(i), the simultaneous equations defeated a large number of candidates. No credit was given for mistakenly using their equation from part (a)(ii) instead of the correct equation for AB.

Q24.

Most candidates scored the mark for the correct printed equation in part (a), but some omitted "= 0" and others made algebraic slips when taking terms from one side of their equation to the other.

In part (b)(i) only the more able candidates were able to obtain the printed inequality using correct algebraic steps. Many began by stating that the discriminant was less than 0, clearly being influenced by the answer. Not all assigned the correct terms to a, b and c in the expression $b^2 - 4ac$ and others made sign errors when removing brackets.

The factorisation was usually correct in part (b)(ii), but many wrote down one of the critical values as $\frac{1}{3}$. Most found critical values and either stopped or immediately tried to write down a solution without any working. Candidates are strongly advised to use a sign diagram or a sketch showing their critical values when solving a quadratic inequality.

Q25.

In part (a) most candidates differentiated the equation of the curve correctly and the majority of these recognised that at the maximum point the value of $\frac{dy}{dx}$ was 0, although some worked with $\frac{d^2y}{dx^2}$ = 0.

In part (b) many candidates had difficulty solving the equation $22.5x^{\frac{1}{2}} - 2.5x^{\frac{3}{2}}$ = 0. There was some invalid squaring of expressions and terms and also it was not uncommon to see $x^{\frac{3}{2}} \div x^{\frac{1}{2}}$ simplified incorrectly either to x^3 or to x^2.

With the equation of the tangent printed in part (d) candidates had to convince examiners that they were finding the value of $\frac{dy}{dx}$ at x = 1 for the gradient of the tangent in part (c). Although not all candidates did this convincingly, the majority did show sufficient evidence to reach the printed equation.

The final part of the question proved to be a good discriminator at the top grade level. Although some excellent solutions were seen these were rare. The most common wrong method was to assume that the length of RM was either equal to the length of PM or equal to half the length of PM. Unfortunately a minority of candidates who produced better solutions which started with the correct equation '162 = 20x – 6' failed to score the final mark due to an inability to solve this equation correctly.

Q26.

In part (a), it was only necessary to complete the square for the y-terms. As a result, there were probably fewer errors this year expressing the left-hand side of the equation of the circle as $(y – 5)^2$. However, the right hand side was often written as $\sqrt{5}$, –5 or –45 instead of 5.

In part (b), quite a number who had the correct circle equation in part (a) wrote the coordinates of the centre as (5, 0) or (0, –5). Generous follow through marks were awarded for the radius provided the right-hand side of the equation had a positive value. The wording in the question reassured most, though, that the radius was $\sqrt{5}$.

In part (c)(i), those with poor algebraic skills, often writing $2x^2$ instead of $(2x)^2$, struggled to establish the given quadratic equation. Also, quite a few made errors in their working but miraculously wrote down the given equation on their final line. A surprising number derived an equation in y. Quite a few simply solved the given quadratic equation in this part and thus failed to show an understanding of what was required.

In part (c)(ii), it was necessary to state that the equation had a repeated root of x = 2, or to use the zero value of the discriminant to show that the equation had equal roots, and hence to conclude that the line was a tangent to the circle.

In part (d), far too many simply substituted the coordinates of the point Q into the equation of the circle obtaining a nonsensical statement such as "–3 = 0 so the point lies inside the circle". It was necessary to see that the distance CQ was being calculated and then concluded that this distance was less than the radius of the circle, and hence the point Q must lie inside the circle.

Q27.

In part (a), some weaker candidates did not realise how to derive the given equation, and others made algebraic slips when proving the printed result, or failed to write "= 0".

In part (b), the condition for two distinct points of intersection required candidates to use the condition that $b^2 - 4ac > 0$ at any early stage of their argument. Those who simply wrote "> 0" on their final line of working, without any previous reference to the discriminant being positive, failed to convince the examiners that they deserved full marks.

In part (c), quite a number were unable to factorise the quadratic correctly and many resorted to using the quadratic equation formula to find the critical values. Where this was done correctly but left in surd form, it was given due credit except for the final mark. Very able candidates can write down the answer to the inequality once they have factorised the quadratic but far too many guessed at answers and an approach using a sign diagram or sketch is recommended. Candidates also need to realise that the final form of the answer cannot be written as $\frac{14}{9} < k < -2$.

Q28.

In part (a), there were some very good sketches, although not all indicated the values of the intercepts as requested. The most common omission was the x-intercept of the line, ($\frac{1}{3}$, 0), possibly because it involved a fraction. The other common error was the omission of the y-intercept of the curve. Many wrongly assumed (0, –3) to be the minimum point of the curve. A few drew the parabola upside down but were given some credit if they marked the correct intercepts on the x-axis. Many who plotted points produced a J-shape that failed to cross the x-axis twice.

In part (b), there were many satisfactory solutions, although quite a few candidates presented their proof in part (a) or (c) of the question. Some omitted "= 0" even though the equation was printed in the question paper. A few candidates guessed two values, possibly from their graph, and merely checked them in the given equation and scored no marks. Many weaker candidates made no attempt at this part.

Part (c) was answered well, although a few stopped after finding only the x-coordinates. Some wrongly substituted these values into $x^2 - x - 2$, instead of the equation of the line or curve, and thus failed to find the y-coordinates of the points of intersection. Quite a few picked out at least one of the pairs of coordinates, possibly using their graphs, and this was given credit.

Q29.

Some candidates did not attempt part (a). However, those who could square $9 - 3x$ usually earned the marks. A surprising number substituted for both x and y and most were unable to complete the solution successfully.

In part (b)(i), the differentiation was carried out very well by most candidates. Errors in the value of k occurred due to taking out a common factor either before or after differentiating, so values of k such as 3 and 9 were common. A few weak candidates did not recognise that they had to differentiate V here, and generally were unable to proceed through the rest of the question.

In part (b)(ii), the factorisation was usually carried out successfully, and candidates who had not found the correct value of k were still able to earn these marks.

The most common error in part (c) was differentiating $x^2 - 4x + 3$, rather than $27(x^2 - 4x + 3)$, and this approach earned a method mark only.

Credit was given in part (d)(i) for substituting their values of x into their second derivative and so most candidates earned this mark, provided they had two values of x.

In part (d)(ii), most realised that $x = 1$ needed to be selected since $\frac{d^2y}{dx^2}$ was negative when $x = 1$.

In part (d)(iii), not all substituted $x = 1$ into the equation for V. Those who did often made arithmetic errors. Some gave the value of y when $x = 1$ and others gave the answer as -54, the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

Q30.

Part (a)(i) It was disappointing to see many candidates unable to rearrange $3x + 5y = 8$ to make y the subject in order to find the gradient. Some were successful in finding a second point on the line such as (1, 1) and then using the coordinates of A to find the gradient of AB.

Part (a)(ii) Most candidates knew how to find the gradient of a perpendicular line, but those using $y = mx + c$ made more arithmetic slips than those using the more appropriate form
$y - y_1 = m(x - x_1)$.

Part (b) Apart from those who used the wrong pair of equations, this part was usually answered correctly.

Part (c) Although this part was meant to be challenging, there were many successful attempts, particularly by those who used a sketch and reasoned on a 3, 4, 5 triangle. It had been intended that candidates would have formed an equation such as $16 + (k + 2)^2 = 25$, but more commonly something such as $16 + y^2 = 25$ was seen, resulting in the incorrect values of ± 3.

Q31.

Part (a) The + $2x$ term was ignored by many who wrote the left hand side of the circle equation as $(x - 1)^2 + (y - 6)$ but most candidates were able to complete the square correctly. The right hand side was often seen as 49 and since this was a perfect square it did not cause candidates to doubt their poor arithmetic.

Part (b) Many who had the correct circle equation in part (a) wrote the coordinates of the centre with incorrect signs. Generous follow through marks were awarded in this part provided the right hand side of the equation had a positive value.

Part (c) Almost all candidates reasoned correctly by considering the y-coordinate of the centre and the radius of the circle, although a number were successful in showing that the quadratic resulting from substituting $y = 0$ into the circle equation does not have real roots. Some simply drew a diagram and this alone was not regarded as sufficient to prove that the circle did not intersect the x-axis. Others using an algebraic approach found a quadratic that they said did not factorise and concluded incorrectly that the equation had no real roots.

Part (d) The algebra proved too difficult for the weaker candidates and many who had shown good algebraic skills rather casually forgot to include "= 0" on their final line of working. Sadly, many were unable to factorise the quadratic or wrote the coordinates of Q as (–5, 2). It was good; however, to see more candidates being able to find the correct mid-point, where in previous years too many had found the difference of the coordinates before dividing by 2.

Q32.

Part (a)(i) The completion of the square was done successfully by most candidates, although occasionally + 44 was seen instead of – 6 for q.

Part (a)(ii) Most candidates were able to write down the correct minimum point, although some wrote (5, – 6) as the vertex. A few chose to use differentiation but often made arithmetic slips in finding the coordinates of the stationary point.

Part (a)(iii) Although there were correct answers for the equation, the term " line of symmetry" was not well understood by many; typical wrong answers were $y = -6$, the y-axis and even
$y = -x^2 + 10x + 19$ or other quadratic curves.

Part (a)(iv) The more able candidates earned full marks here. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch. The most common (but incorrect) vector stated was
$\begin{bmatrix} 10 \\ 19 \end{bmatrix}$.

Part (b) There were a number of complete correct solutions here. The errors that did occur usually stemmed from sign slips in rearranging the equations. Some candidates found the x-coordinates and made no attempt at the y-coordinates. A few candidates wrote down the coordinates of at least one point without any working.

Q33.

(a) (i) Most candidates were familiar with the idea of "completing the square" and answered this part satisfactorily. There were occasional sign errors and + 9 – 4 was not always evaluated correctly.

(ii) There were several correct answers although some wrote (5,–2) instead of (5,2). Some did not recognise the link between parts (i) and (ii) and chose to differentiate instead. This was a satisfactory, though more time-consuming, alternative method. Some earned no marks here as they wrote comments such as "5 is the minimum", with no link to the y-coordinate being 5.

(b) (i) This simple proof was usually well done. Occasionally the mark was lost due to the omission of " = 0"..

(ii) Many scored full marks here. Most factorised the equation and obtained the correct x-values. Some made no further progress, while a few substituted into the given quadratic equation and obtained y = 0, instead of using the equation of the line or curve to find the values of y. It was encouraging to see many factorising the quadratic correctly. Those who used the quadratic equation formula or completion of the square often made more errors than those who factorised.

Q34.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting x = 0 to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P. A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle then using the cosine rule to find angle OPA then using the formula $\frac{1}{2}ab\sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more directly by using $\frac{1}{2}OA \times |y_P|$.

Q35.

Part (a)(i) Although most obtained the correct gradient, some omitted the negative sign (particularly those who relied on a sketch for their evaluation) and some had a fraction with the change in x as the numerator which immediately scored no marks. Quite a few found mid-points (possibly since that had appeared on previous examinations) and others added the coordinates instead of finding the differences in their quotient expression for the gradient.

Part (a)(ii) The use of their gradient to obtain the given equation was the most successful method. Those using $y = mx + c$ had a tendency to introduce a new 'c' by doubling both sides but then substituted their value back into the original equation. The most successful candidates used the formula $y - y_1 = m(x - x_1)$. Some re-arranged the given equation to check the gradient then checked one set of co-ordinates; others checked two points and indicated that a straight line has the form $ax + by = c$.

Part (b) Those using substitution often began by using an incorrect rearrangement of one of the equations. If they attempted elimination, sometimes only part of an equation was multiplied by the appropriate constant. Many added the equations instead of subtracting. Of those who wrote 14y = –7, just as many obtained an incorrect answer of $y = -2$ as the correct answer of $y = -1/2$.

Part (c) The condition for perpendicularity was generally known but some were unable to evaluate –1 divided by – 1.5. A few omitted the – sign while some referred to the equation $3x + 2y = 17$ and gave a gradient of 1/3. Once again, those determined to use $y = mx + c$ often made errors in the constant due to the fractional coefficient of x. Quite a few did not use the point s as instructed, choosing to use the point C instead.